

2520030453

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Section-6

3.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 practical3.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pid = fork();

    if (pid == 0)
    {
        printf("Child: PID=%d PPID=%d\n", getpid(), getppid());
        printf("Child: Running\n");

        printf("Child: Waiting\n");
        sleep(10);

        printf("Child: Running again\n");
        printf("Child: Terminated\n");
    }
    else
    {
        printf("Parent: PID=%d PPID=%d\n", getpid(), getppid());
        printf("Parent: Waiting for child\n");

        wait(NULL);

        printf("Parent: Child terminated\n");
    }

    return 0;
}
```

```
siri_chandana@LAPTOP-MMV × + ▾
siri_chandana@LAPTOP-MMV:~$ nano practical3.c
siri_chandana@LAPTOP-MMV:~$ gcc practical3.c -o shell
siri_chandana@LAPTOP-MMV:~$ ./shell
Parent: PID=626 PPID=360
Parent: Waiting for child
Child: PID=627 PPID=626
Child: Running
Child: Waiting
```

4.

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 practical4.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i, pid;

    for (i = 0; i < 3; i++)
    {
        pid = fork();

        if (pid == 0)
        {
            printf("Child %d: PID=%d\n", i + 1, getpid());
            sleep(i + 1);
            return 0;
        }
    }

    for (i = 0; i < 3; i++)
        printf("Parent collected PID=%d\n", wait(NULL));

    return 0;
}
```

```
siri_chandana@LAPTOP-MMV × + v
siri_chandana@LAPTOP-MMVMUDEP:~$ nano practical3.c
siri_chandana@LAPTOP-MMVMUDEP:~$ gcc practical3.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~$ ./shell
Parent: PID=626 PPID=360
Parent: Waiting for child
Child: PID=627 PPID=626
Child: Running
Child: Waiting
Child: Running again
Child: Terminated
Parent: Child terminated
siri_chandana@LAPTOP-MMVMUDEP:~$ nano practical4.c
siri_chandana@LAPTOP-MMVMUDEP:~$ gcc practical4.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~$ ./shell
Child 1: PID=672
Child 2: PID=673
Child 3: PID=674
Parent collected PID=672
Parent collected PID=673
|
```